MAT205: Abstract Algebra II

Algebraic Closure via Zorn’s Lemma

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Part I

Statement and Construction

Motivation: From One Root to Every Root

L12 Part IV: every nonconstant $f \in F[x]$ has *some* root in *some* extension E/F.

But this produces one root at a time; the extension depends on the polynomial.

We want **all** roots of every polynomial in a single ambient field – for splitting fields, Galois groups, factorization.

Pushed to the limit: a field $\overline{F}$ in which every nonconstant $f \in F[x]$ has a root, with no algebraic excess.

Goal of this lecture. Such an $\overline{F}$ exists and is essentially unique. Existence uses Zorn's lemma.

Definition: Algebraically Closed Field

Definition. A field K is **algebraically closed** if every nonconstant polynomial $f \in K[x]$ has a root in K.

Equivalent characterization. Every nonconstant $f \in K[x]$ factors completely into linear factors:

$$f = c \prod_{i=1}^{\deg f} (x - \alpha_i), \qquad c \in K^\times,\ \alpha_i \in K.$$

Example. $\mathbb{C}$ is algebraically closed (Fundamental Theorem of Algebra; classical proof outside this course).

Non-examples. $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{F}_p$ – each has irreducible polynomials of degree ≥ 2.

Definition: Algebraic Closure

Definition. An **algebraic closure** of a field F is a field $\overline{F}$ such that:

1. $F \subseteq \overline{F}$ (extension);
2. $\overline{F}/F$ is an **algebraic extension** (L12 Part VI: Definition of Algebraic Extension);
3. $\overline{F}$ is algebraically closed.

Reading. $\overline{F}$ contains a root of every nonconstant $f \in F[x]$, and nothing transcendental is added beyond what algebraicity forces.

Concrete example. $\overline{\mathbb{Q}} := \{\alpha \in \mathbb{C} : \alpha \text{ algebraic over } \mathbb{Q}\}$ is an algebraic closure of $\mathbb{Q}$. Algebraicity from L12 (Algebraic Elements Form a Subfield). Algebraic closedness: $\mathbb{C}$ is algebraically closed plus transitivity (L12: Transitivity of Algebraicity).

Theorem: Steinitz (Existence and Uniqueness)

Theorem (Steinitz, 1910). Every field F has an algebraic closure $\overline{F}$.

Furthermore, any two algebraic closures of F are isomorphic as extensions of F.

Existence uses **Zorn's lemma** (equivalent to the Axiom of Choice).

Uniqueness uses the **isomorphism-extension theorem**.

Historical note. Steinitz's 1910 paper systematized field theory using transfinite methods – one of the earliest serious uses of choice principles in algebra.

Proof Skeleton: The Zorn Argument

Step 1 (Bound the ambient set). All algebraic extensions of F have cardinality $\leq \max(|F|, \aleph_0)$. Fix an ambient set Ω of this cardinality with $F \subseteq \Omega$.

Step 2 (Define the poset). Let

$$\mathcal{P} = \{\, (K, +_K, \cdot_K) : F \subseteq K \subseteq \Omega,\ K \text{ a field},\ K/F \text{ algebraic} \,\},$$

partially ordered by inclusion.

Step 3 (Apply Zorn's lemma). Every chain has an upper bound (the union – a field, algebraic over F). Zorn yields a maximal element $\overline{F} \in \mathcal{P}$.

Step 4 (Maximal $\Rightarrow$ algebraically closed). If $f \in \overline{F}[x]$ has no root in $\overline{F}$, build $\overline{F}(\alpha) \supsetneq \overline{F}$ with $f(\alpha) = 0$ (L12: Theorem on Existence of a Root). Then α is algebraic over F by transitivity (L12: Transitivity of Algebraicity). Embed into Ω – contradicts maximality.

Conclusion. $\overline{F}$ is algebraically closed and algebraic over F – an algebraic closure of F.